ICDFA / Check Point SecureAcademy

CCSA101 LAB01 Check Point Lab Setup and Administration

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Course	**Introduction to Quantum Security**
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Part A: Prepare VMware and the private network

Step 1: Install the Windows tools

Objective

install VMware Workstation Pro and R82 SmartConsole.

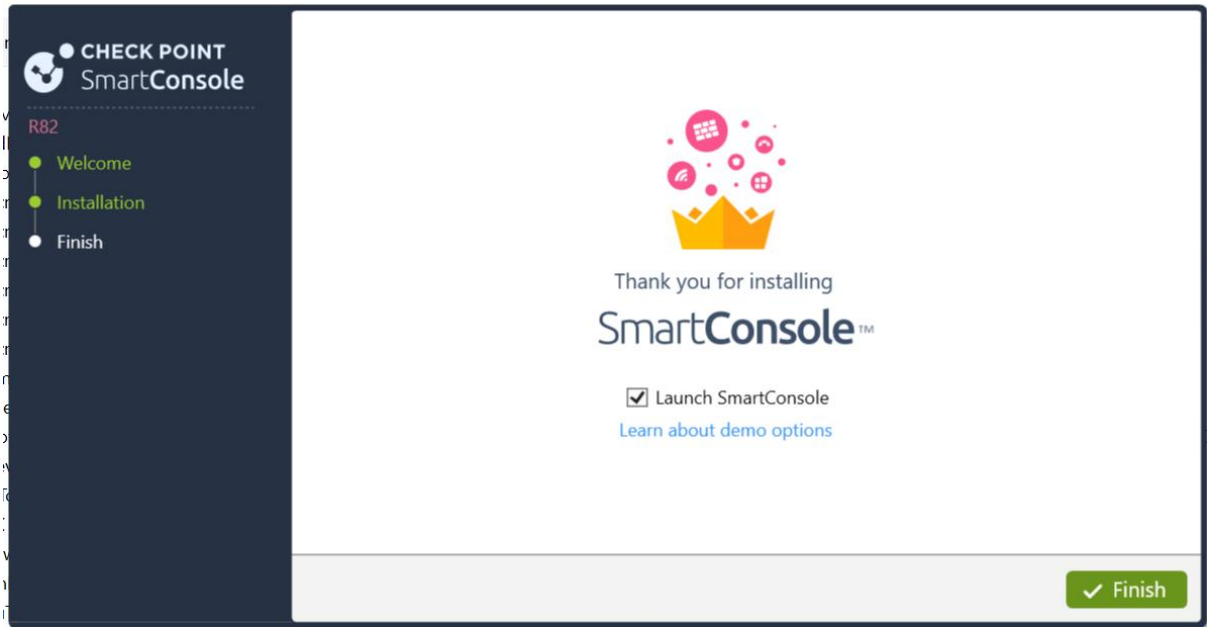


Fig. A1 SmartConsole Interface

Item	Version	Source
SmartConsole	R82 SmartConsole (6.11.0)	checkpoint
VMware Workstation Pro	26.0.0	Getintopc

Explanation

Both required software packages were installed successfully

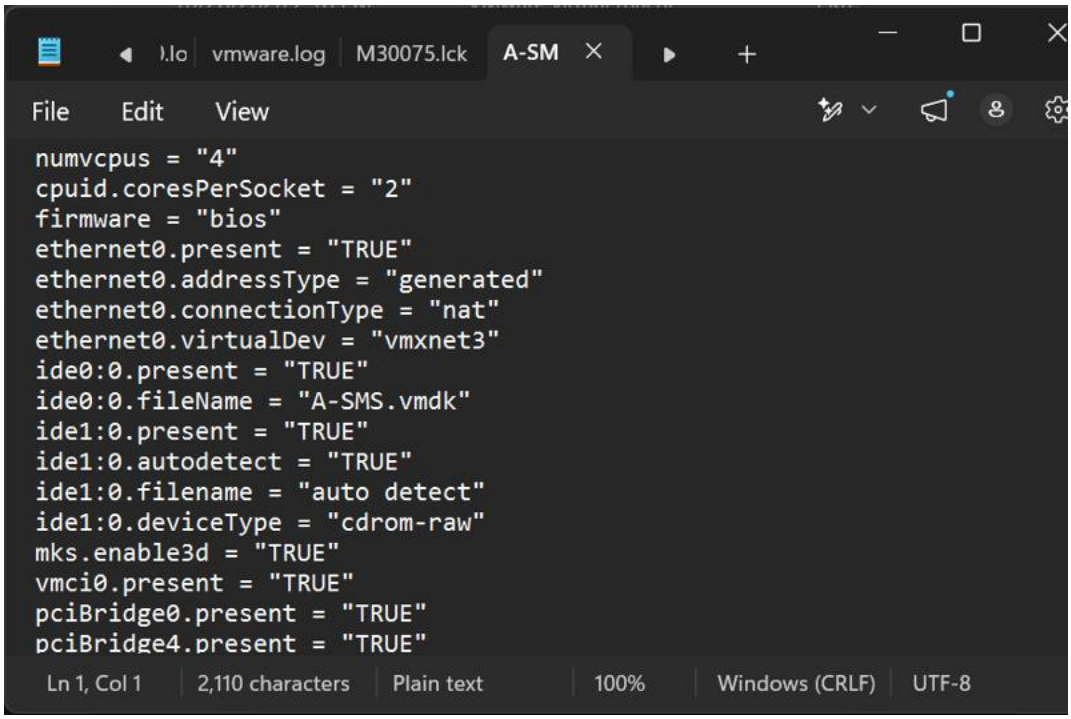
Step 2: Load A SMS and check its adapter

Objective

Verify A-SMS configuration and VMXNET3 adapter.

The A-SMS virtual machine was extracted to a writable local folder and opened in VMware Workstation using the A-SMS.vmx file. The virtual machine name, hardware configuration, and virtual disk were verified while the VM remained powered off.

A backup copy of the .vmx file was created. The VMX file was opened in Notepad and the following entry was confirmed



The file was saved with the **.vmx** extension and reopened in VMware Workstation.

The network adapter was configured as follows:

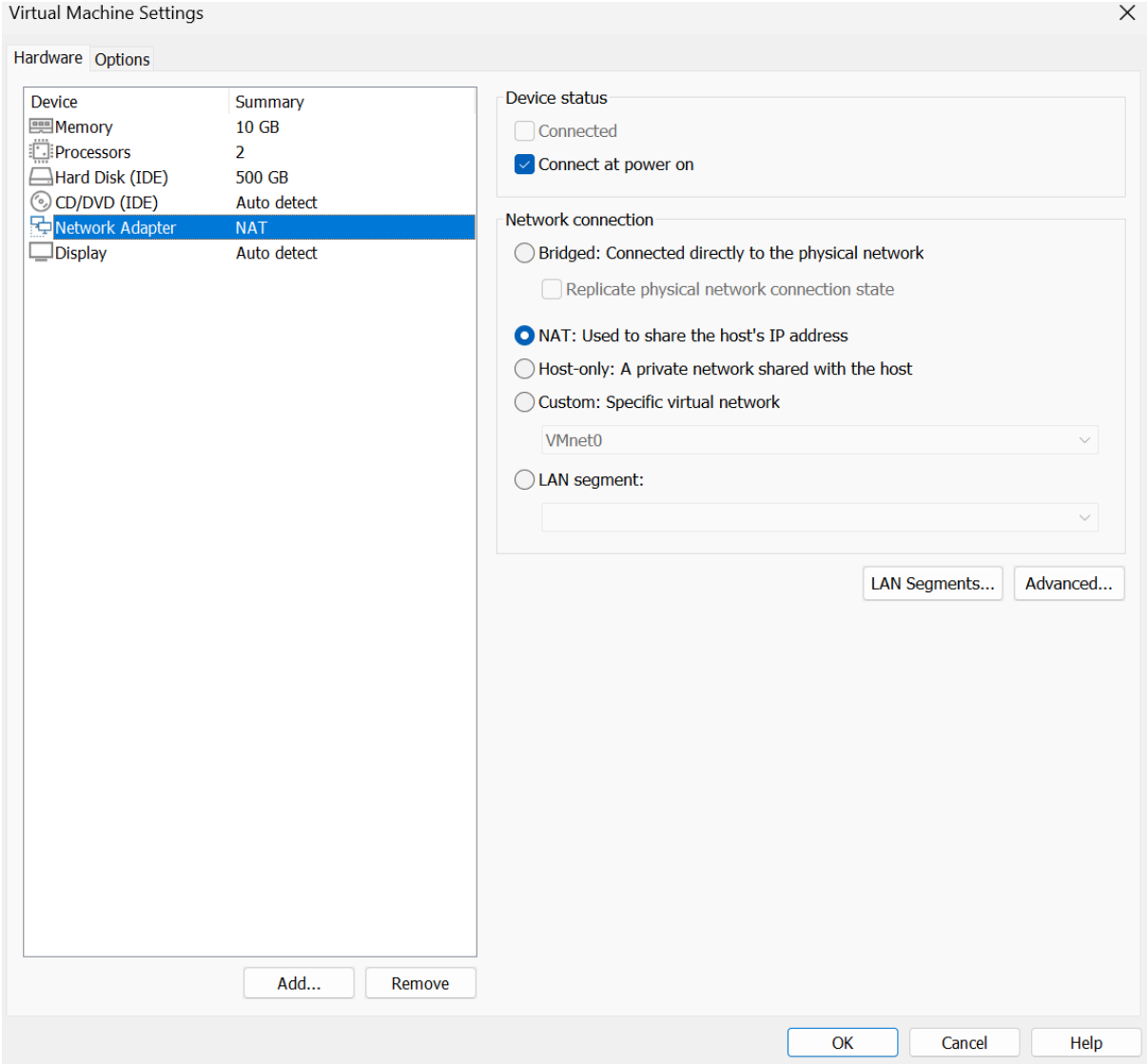


Fig. A2 Network Adapter set to Custom: VMnet8 with Connect at power on enabled.

Interpretation

The VM is configured according to lab requirement

Observation

The A-SMS virtual machine was configured to use a single VMXNET3 network adapter connected to VMnet8. VMXNET3 provides optimized virtual network performance, while VMnet8 supplies a private NAT-based network for the lab environment.

Step 3: Set the VMnet8 network

The VMnet8 virtual network was configured to use NAT networking with the VMware DHCP service enabled. The host virtual adapter remained connected to the VMnet8 network.

Setting	Configured Value
VMnet8 Subnet	10.1.1.0
Subnet Mask	255.255.255.0
NAT Gateway	10.1.1.250
DHCP Address Pool	10.1.1.210 - 10.1.1.240
Windows VMnet8 Adapter	10.1.1.201/24 (No Default Gateway)
A-SMS eth0	Static 10.1.1.101/24

Evidence:

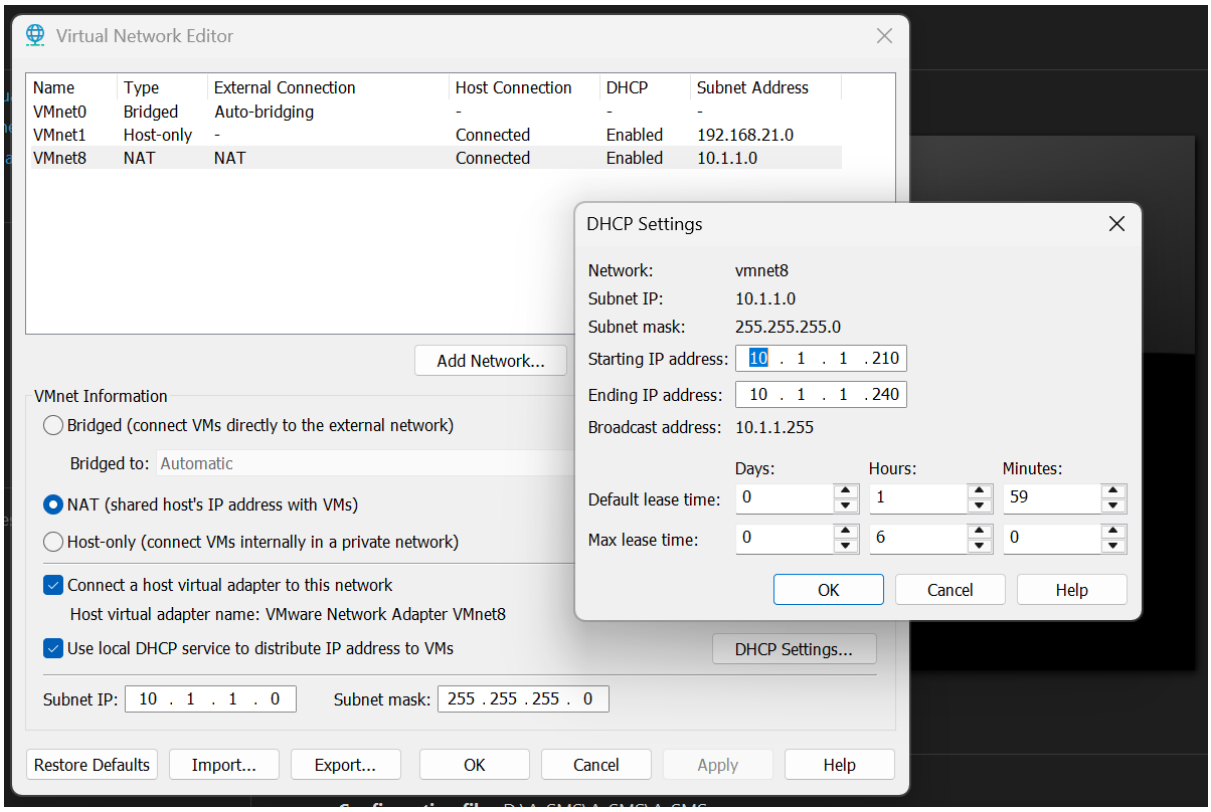


Fig. A3.1 VMnet8 subnet and mask (10.1.1.0 / 255.255.255.0) / DHCP Settings showing pool 10.1.1.210 - 10.1.1.240

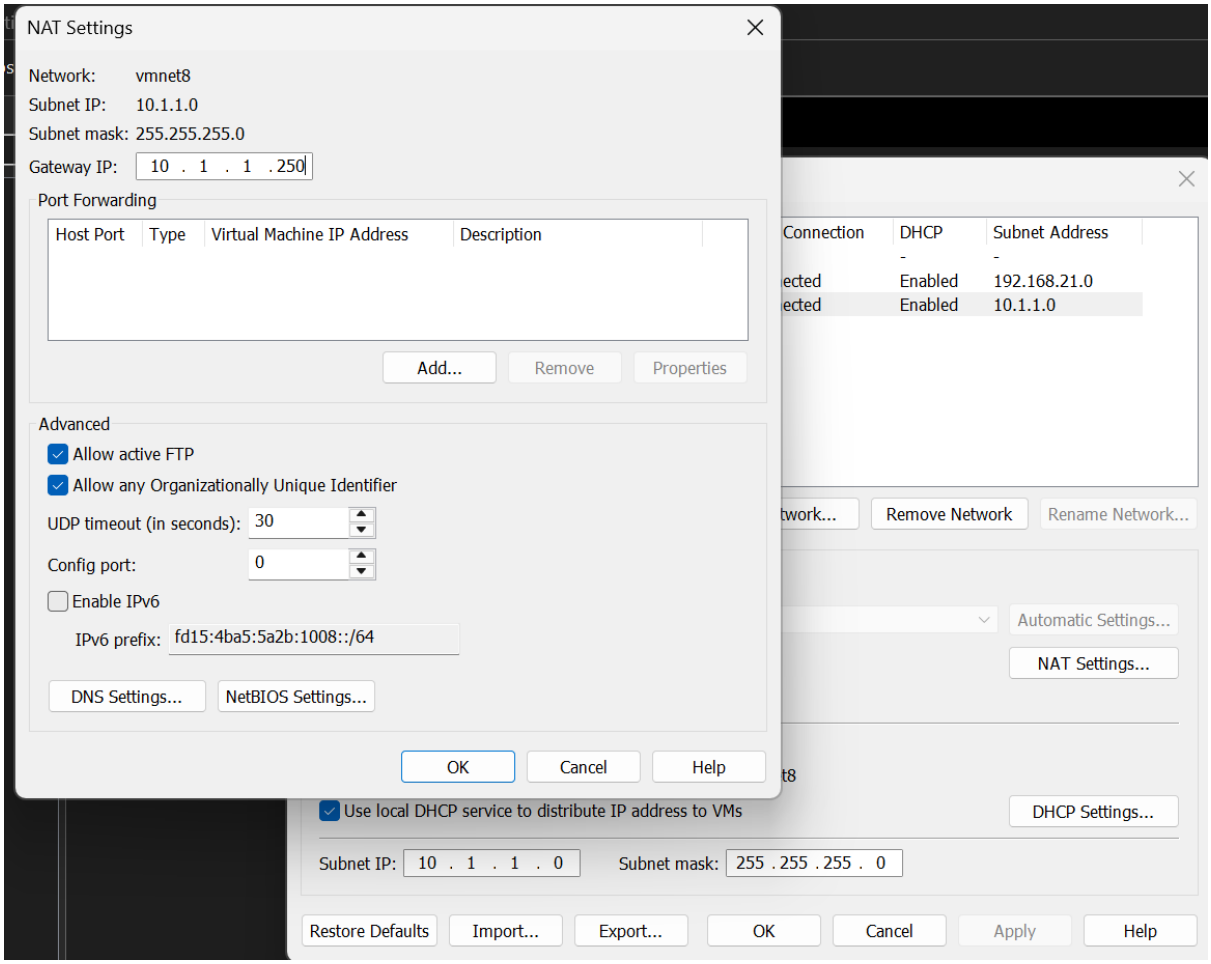


Fig. A3.2 NAT Settings showing gateway 10.1.1.250

Interpretation

VMnet8 provides a private management network independent of the physical Wi-Fi connection.

Part A Set the addresses and verify Gaia

Step 4: Set the Windows VMnet8 address

The VMnet8 adapter was configured with a static IPv4 address using the netsh command. The configuration was applied only to the VMware Network Adapter VMnet8, while the physical Wi-Fi and Ethernet settings remained unchanged

Settings	Value
Adapter	VMware Network Adapter VMnet8
IPv4 Address	10.1.1.201
Subnet Mask	255.255.255.0

Default Gateway	None

```
Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::da5d:e5b8:14e5:699f%7
   IPv4 Address. . . . . . . . . . . : 10.1.1.201
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

Fig. A4 The Command Prompt showing the ipconfig output.

Step 5: Boot A SMS and inspect eth0

Property	Value
Interface	eth0
IPv4 Address	10.1.1.101
Mask Length	255.255.255.0
Administrative State	ON
Link Status	UP
Interface Status	UP

```
A-SMS> show inte
interface  - Show a specific interface's configurations
interfaces - Lists all interfaces
A-SMS> show interface e
A-SMS> show interface eth0
state on
mac-addr 00:0c:29:6f:3b:e9
type ethernet
link-state link up
mtu 1500
auto-negotiation off
speed 10G
ipv6-autoconfig Not configured
monitor-mode Not configured
duplex full
link-speed 10G/full
comments
ipv4-address 10.1.1.101/24
ipv6-address Not Configured
ipv6-local-link-address Not Configured

Statistics:
TX bytes:14143922 packets:11998 errors:0 dropped:0 overruns:0 carrier:0
RX bytes:838486 packets:5962 errors:0 dropped:0 overruns:0 frame:0
A-SMS> _
```

Fig:A5 interface eth0 of A-SMS

Step 6: Check access from Windows

Command used on windows terminal

ping 10.1.1.101

```
C:\Users\Ethick_Fahad>ping 10.1.1.101

Pinging 10.1.1.101 with 32 bytes of data:
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64

Ping statistics for 10.1.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig. A6.1 The ping 10.1.1.101 result

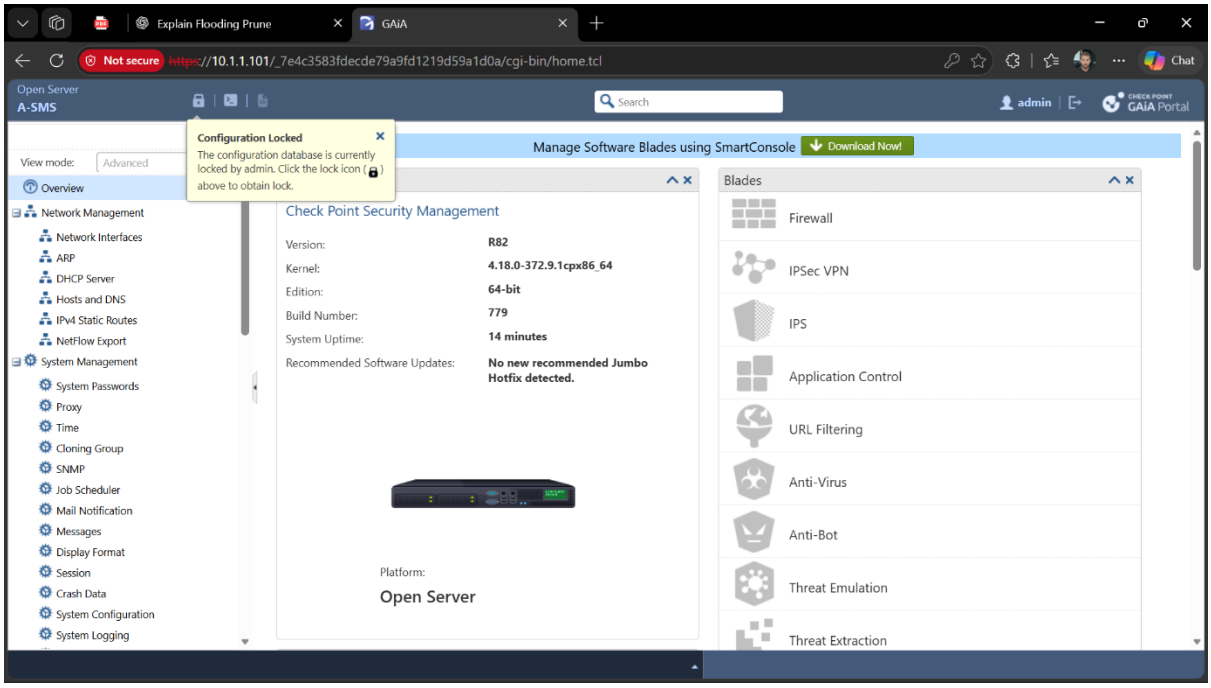


Fig A6 The Gaia login page at **https://10.1.1.101**.

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Part A Connect SmartConsole

Step 7 Verify the server and sign in

SmartConsole

The R82 SmartConsole client was launched and configured to connect to the management server at 10.1.1.101. Authentication was completed using the supplied SmartConsole account.

ICA Fingerprint Verification

To verify the identity of the management server, the ICA fingerprint displayed by SmartConsole was compared with the fingerprint retrieved directly from the trusted A-SMS VMware console.

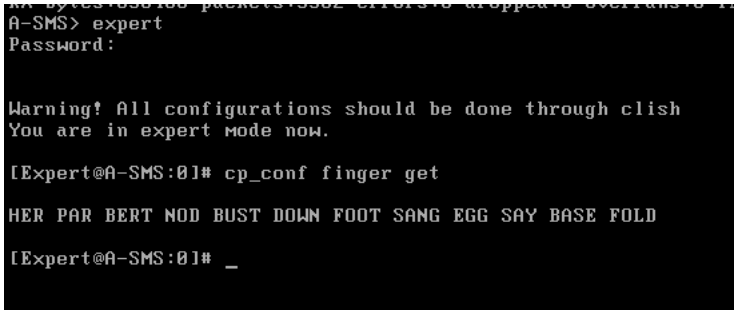


Fig A7.1 The cp_conf finger get output from the A-SMS console.

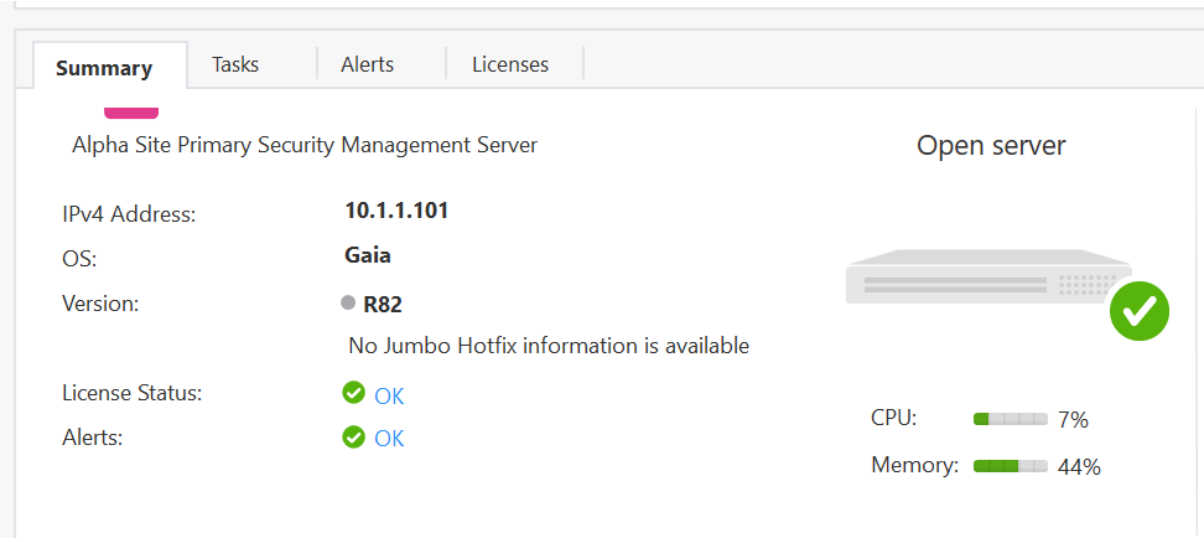


Fig. A7.2 The A-SMS object in Gateways & Servers.

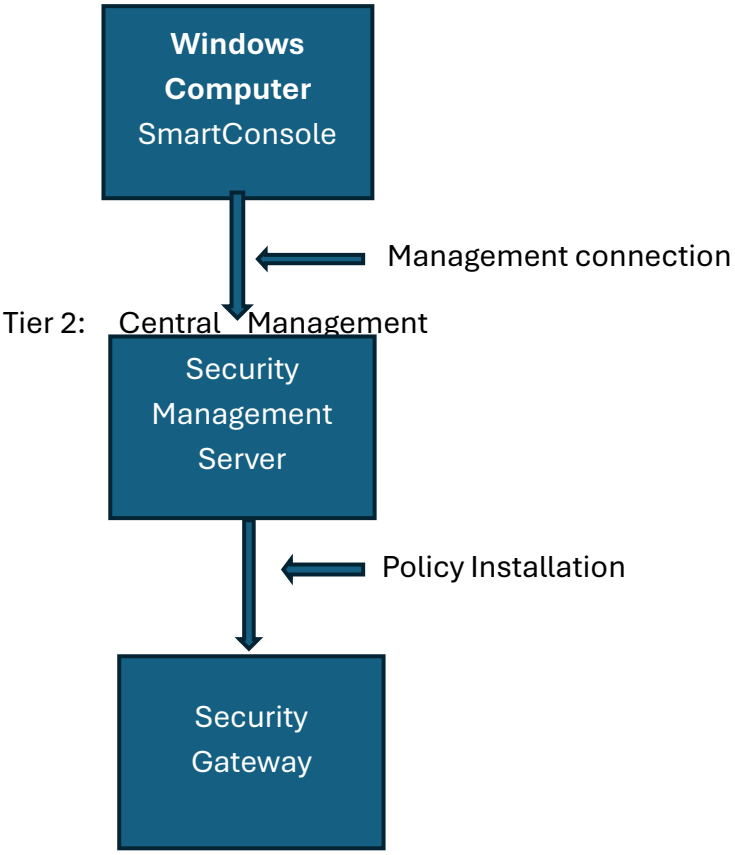
Observation

SmartConsole successfully authenticated to A-SMS (10.1.1.101). The ICA fingerprint presented during the first connection was verified using cp_conf finger get from the trusted VMware console before the connection was accepted. This confirmed the authenticity of the management server.

Part B: Explain the Check Point architecture

B1: Explain the architecture

Tier 1: Management interface



Role of Each Component

SmartConsole

- The administrator's management application installed on Windows.
- Used to create security policies, manage objects, monitor logs, and administer Check Point devices.

Security Management Server (A-SMS)

- Stores security policies, network objects, users, and configuration data.
- Processes administrator changes made through SmartConsole.
- Manages policy publication and deployment.

Security Gateway

- Enforces the security policy on network traffic.

- Inspects, allows, or blocks traffic according to the installed policy.
- Generates logs and security events.

Security policies and network object definitions are maintained in the Security Management Server (SMS) database.
Administrators use SmartConsole to create and edit these configurations, but the SMS stores and manages them centrally.

Changing a policy and enforcing a policy are two different processes in Check Point architecture.
Changing a policy means creating, editing or modifying security rules and network objects through SmartConsole. The changes are saved in the Security Management Server after the administrator publishes the session.
Enforcing a policy means installing the published security policy from the SMS onto the Security Gateway. Once successfully installed, the Security Gateway applies the rules when inspecting network traffic.

B2: Explain management services and deployment

Explain three Security Management Server functions;
a, Logging and Monitoring:
The SMS collects and manages security logs generated by Security Gateways. It allows administrators to monitor network activities, review security events, identify suspicious traffic and generate reports. This helps administrators understand what is happening within the network and respond to possible security threats.

b, Security Policy Management
The SMS manages the security policies used to control network traffic. It allows administrators to create, modify and manage firewall rules that determine which traffic should be allowed or denied. It also installs these policies on Security Gateways for enforcement.

c, Network Object Management:
The SMS maintains and stores network object definitions such as hosts, networks, services and Security Gateways. These objects make it easier for administrators to create and manage security rules without repeatedly entering IP addresses and other details.

Comparison Between On-Premises Management and Smart-1 Cloud.

On-Premises Management	Smart-1 Cloud
Management server is hosted and maintained by the organization.	Management platform is hosted and maintained by Check Point.
Organization is responsible for hardware, backups, updates, and maintenance.	Check Point manages infrastructure maintenance and platform updates.
Greater direct control over the management environment.	Reduced need for on-site hardware and management effort.
Requires hardware resources and administrative overhead.	Requires internet connectivity to access the cloud management service.

Recommendation: For an organisation that wants to reduce on-premises hardware maintenance, Smart-1 Cloud is the recommended option.
Because Check Point provides the management service through the cloud, reducing the need for the organisation to purchase, install and maintain a dedicated management server. It also provides centralised security policy management, log analysis and reporting.

Trade-off: Smart-1 Cloud depends on reliable internet connectivity and the availability of the cloud service. The organisation also has less direct control over the underlying management infrastructure compared to an on-premises deployment.

Conclusion
The Security Management Server provides policy management, object management, and log management functions. While on-premises deployment offers greater control, Smart-1 Cloud reduces hardware maintenance and administrative effort, making it a suitable choice for organizations seeking a simpler management solution.

B3 Choose the correct administration interface

Task	**Best Tool**	**Justification**
Inspect eth0	Gaia Clish	Gaia Clish provides commands such as show interface eth0 to display interface details, IP addresses, and subnet masks.

Review an Access Control rule	SmartConsole	Access Control policies and rules are created, viewed, and managed through SmartConsole
Inspect operating-system resource use	Expert Mode	Expert Mode provides access to Linux commands such as top, free, and df for monitoring CPU, memory, and disk usage.
View security logs	SmartConsole	The Logs & Events section of SmartConsole allows administrators to view and analyze security logs and events.

Difference Between Gaia Clish, Expert Mode, and SmartConsole Command Line Panel

Gaia Clish

- Check Point's administrative command shell.
- Used mainly for network and system configuration.
- Provides structured commands and prevents many configuration errors.

Expert Mode

- Gives direct access to the underlying Linux operating system.
- Used for advanced troubleshooting, diagnostics, and resource monitoring.
- Supports standard Linux commands.

SmartConsole Command Line Panel

- A command-line interface accessed through SmartConsole.
- Allows administrators to run commands remotely on managed Check Point devices.
- Does not require a separate SSH login to the gateway.

B4: Explain security features and policy handling

1. **Network Security**
   A company wants to control which traffic can enter or leave its network. Network Security enforces firewall rules to allow legitimate traffic and block unauthorized access, protecting internal systems from attacks.
2. **Threat Prevention**
   A company wants to prevent malware, exploits, and other cyber threats from reaching users and servers. Threat Prevention inspects traffic and automatically detects and blocks malicious content before it can cause damage.

Why Publishing and Installing Policy Are Separate Actions

Publishing a Management Session

- Saves the administrator's changes to the Security Management Server database.
- Makes the changes visible to other administrators.
- Does not affect live traffic on any gateway.

Installing Policy on a Gateway

- Sends the published policy to the Security Gateway.
- Causes the gateway to begin enforcing the updated rules on network traffic.
- Applies the changes to the production environment.

B5. Interpret Your Own Lab Evidence

The VMnet8 addresses can remain fixed when Wi-Fi changes because VMnet8 is a VMware virtual network that operates independently of the physical network adapter. The Windows VMnet8 adapter (10.1.1.201) and the A-SMS interface (10.1.1.101) stay on the private 10.1.1.0/24 network regardless of changes to the laptop's Wi-Fi connection.

Enabled VMware DHCP does not change the A-SMS address because A-SMS uses a manually configured static IP address (10.1.1.101/24). VMware DHCP only allocates addresses to devices configured for automatic addressing. Since the A-SMS address is outside the DHCP pool and statically assigned, it remains unchanged.

A successful ping proves basic IP connectivity between the Windows host and A-SMS. A successful SmartConsole login proves more than network connectivity because it confirms that the Security Management Server is reachable, management services are running, and authentication is working correctly

```
C:\Users\Ethick_Fahad>ping 10.1.1.101

Pinging 10.1.1.101 with 32 bytes of data:
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64
Reply from 10.1.1.101: bytes=32 time<1ms TTL=64

Ping statistics for 10.1.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

An unavailable gateway object does not automatically indicate a fault. In the lab environment, some gateways may appear unavailable simply because they are powered off or not deployed. Additional evidence, such as connectivity tests and status information, is required before diagnosing a problem.
Similarly, an empty log view does not prove that no traffic or security events occurred. The result may be caused by log filters, time-range selection, logging configuration, communication issues, or unavailable log sources. Further investigation is required before concluding that a fault exists.

Conclusion:
The lab demonstrated that VMware private networking operates independently of Wi-Fi, static addresses are unaffected by DHCP, and different management tools provide different levels of connectivity evidence. Status indicators and log results should always be supported by additional evidence before diagnosing a fault.

Part C Explore Gaia Portal

C1: Inspect the system overview

ITEM	VALUE
Hostname	A-SMS
Product release	R82
Build Information	779
Uptime	1hr 27mins
CPU and Memory used	10% and 45 %

Why resource readings can change:
CPU and memory usage change continuously because the system processes different amounts of network traffic, user activity, security inspections, logging, and background services. As workload increases or decreases, resource utilization changes accordingly.
Fig. G1 image of system overview through Gaia portal

Observation
The Gaia System Overview page provided information about the A-SMS management server, including the hostname, software version, build information, uptime, and current resource usage. The system was running Check Point R82 Build 779, and the resource statistics reflected the current workload at the time of observation.

C2: Compare the navigation modes

Basic View

- Provides simplified navigation with only the most commonly used administration tasks.

- Easier for new administrators and daily management.

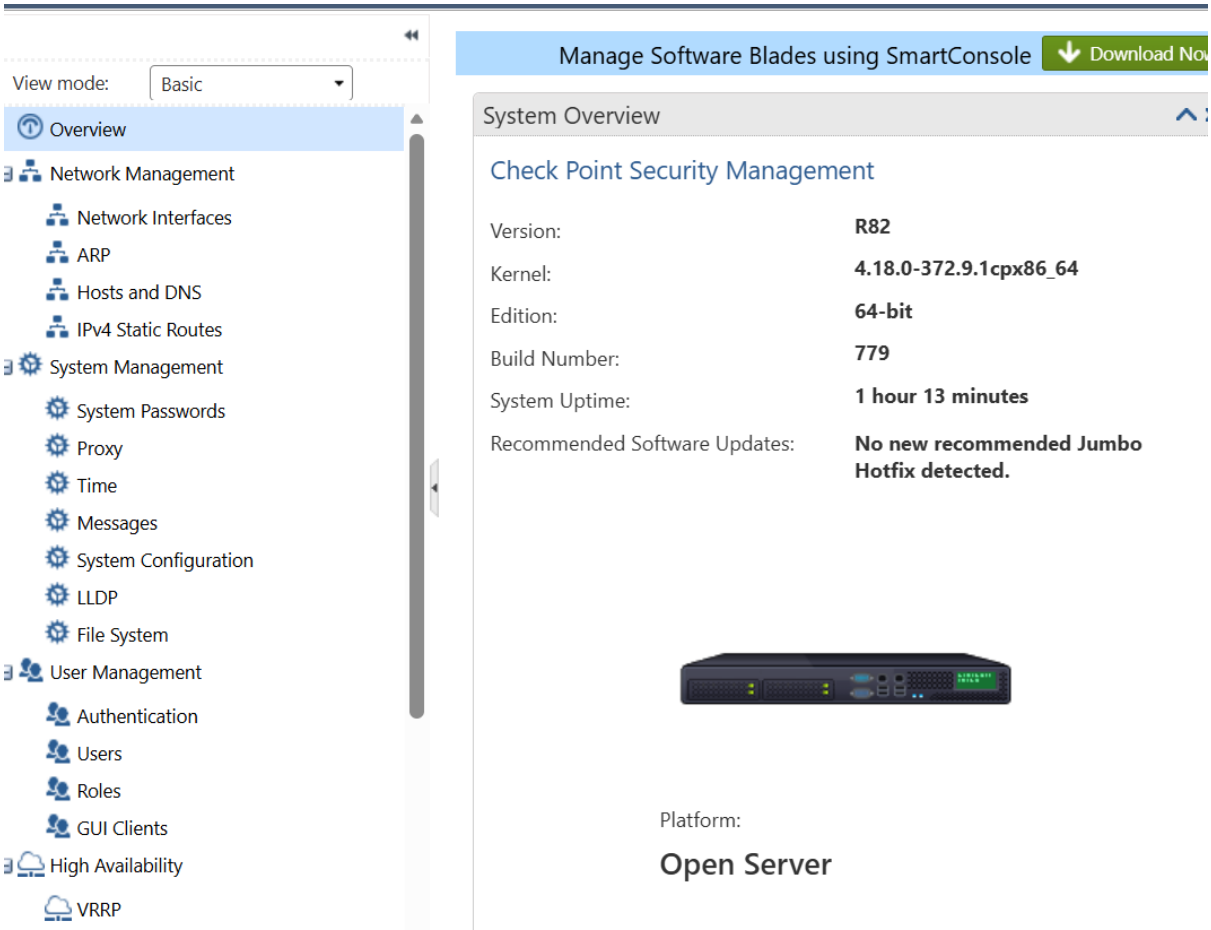


Fig. Basic view mode

Advanced View

- Provides access to additional configuration and monitoring options.

- Intended for more detailed administration and troubleshooting.

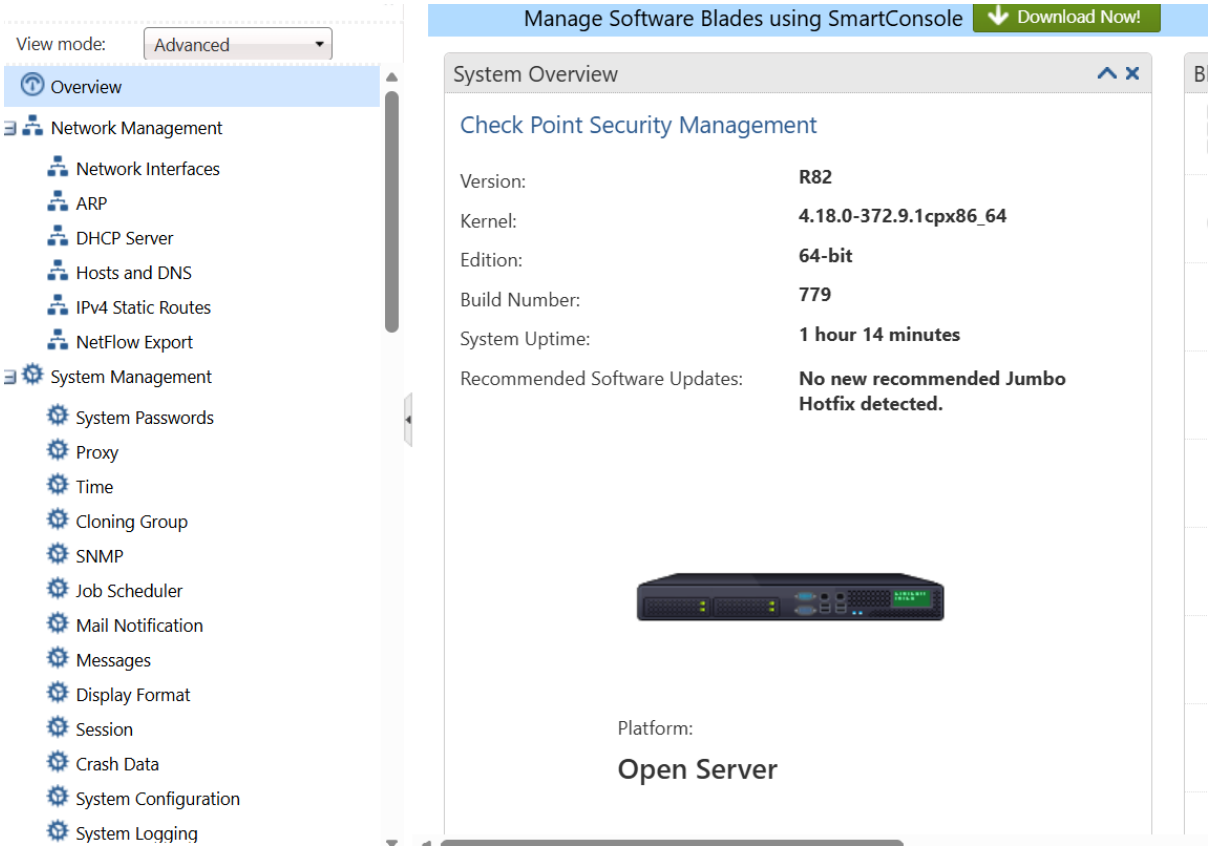


Fig. Advance View Mode

Navigations Area

1. System Management: Configure system settings such as hostname, time, and updates.
2. Network Management: Configure interfaces, routes, and network setting

Option Available Only in Advanced View.

NetFlow Export: is available only in Advanced View. It is used to export network traffic flow information to a NetFlow collector for traffic analysis and monitoring.

Interface Limitation.

Available menus may vary depending on the Gaia version and user permissions. In this lab, NetFlow Export was visible only in Advanced View.

C3 Inspect the management interface

Observation

ITEMS	VALUE
Interface	eth0
IPv4 Address	10.1.1.101
Subnet Mask	255.255.255.0
Administrative State	Up
MAC Address	00:0c:29:6f:3b:e9
Link Status	Up

Evidence.

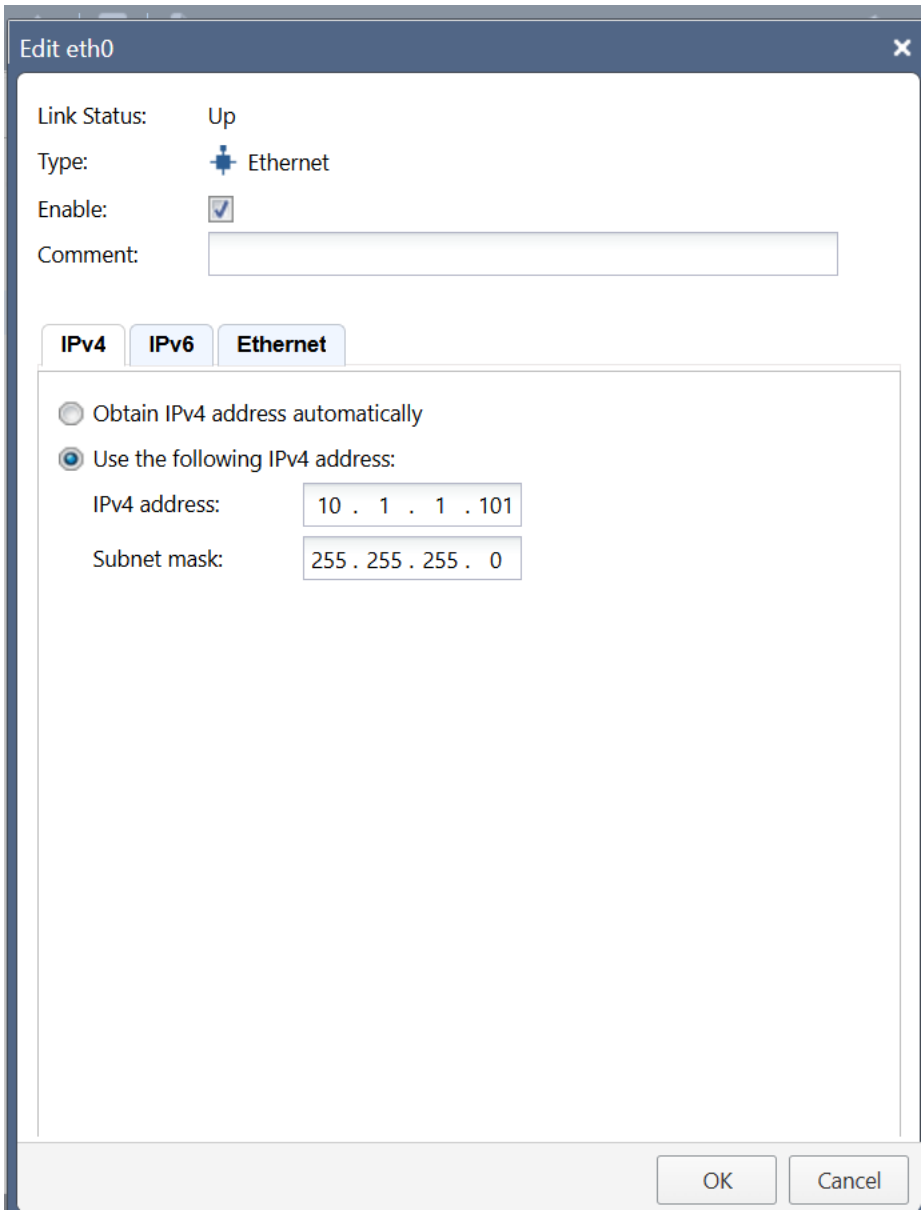


Fig. G3 eth0 interface on Gaia portal

C4 Inspect routes and DNS

Observation

ITEM	Value
Default Gateway	10.1.1.1
DNS Server	168.63.129.16

Difference Between a Default Gateway and a DNS Resolver

- A default gateway is the router that forwards traffic to networks outside the local subnet. It is used when the destination is not on the local network.
- A DNS resolver translates domain names (such as www.google.com) into IP addresses so that devices can locate servers on a network.

Evidence

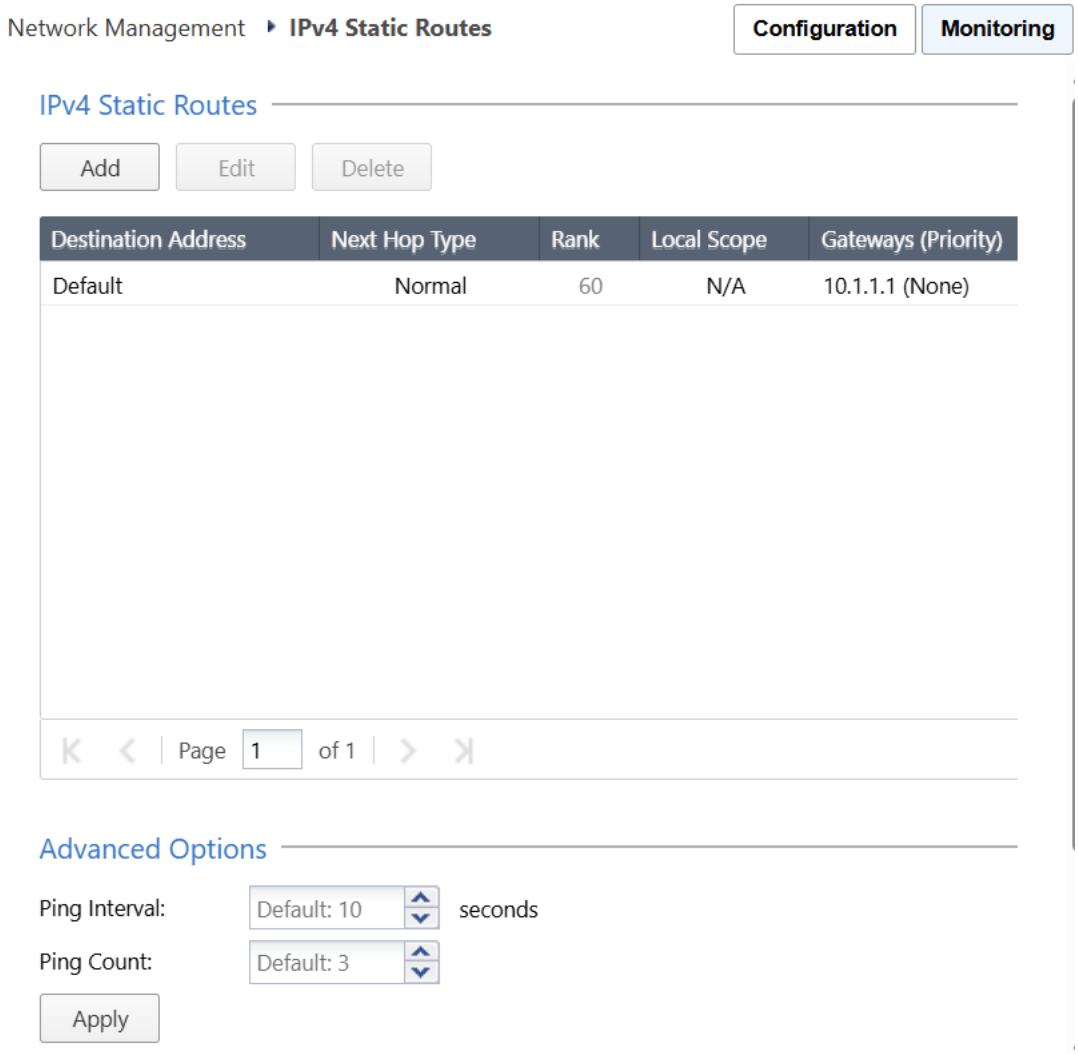


Fig. G IPv4 Routing/Static Routes page.

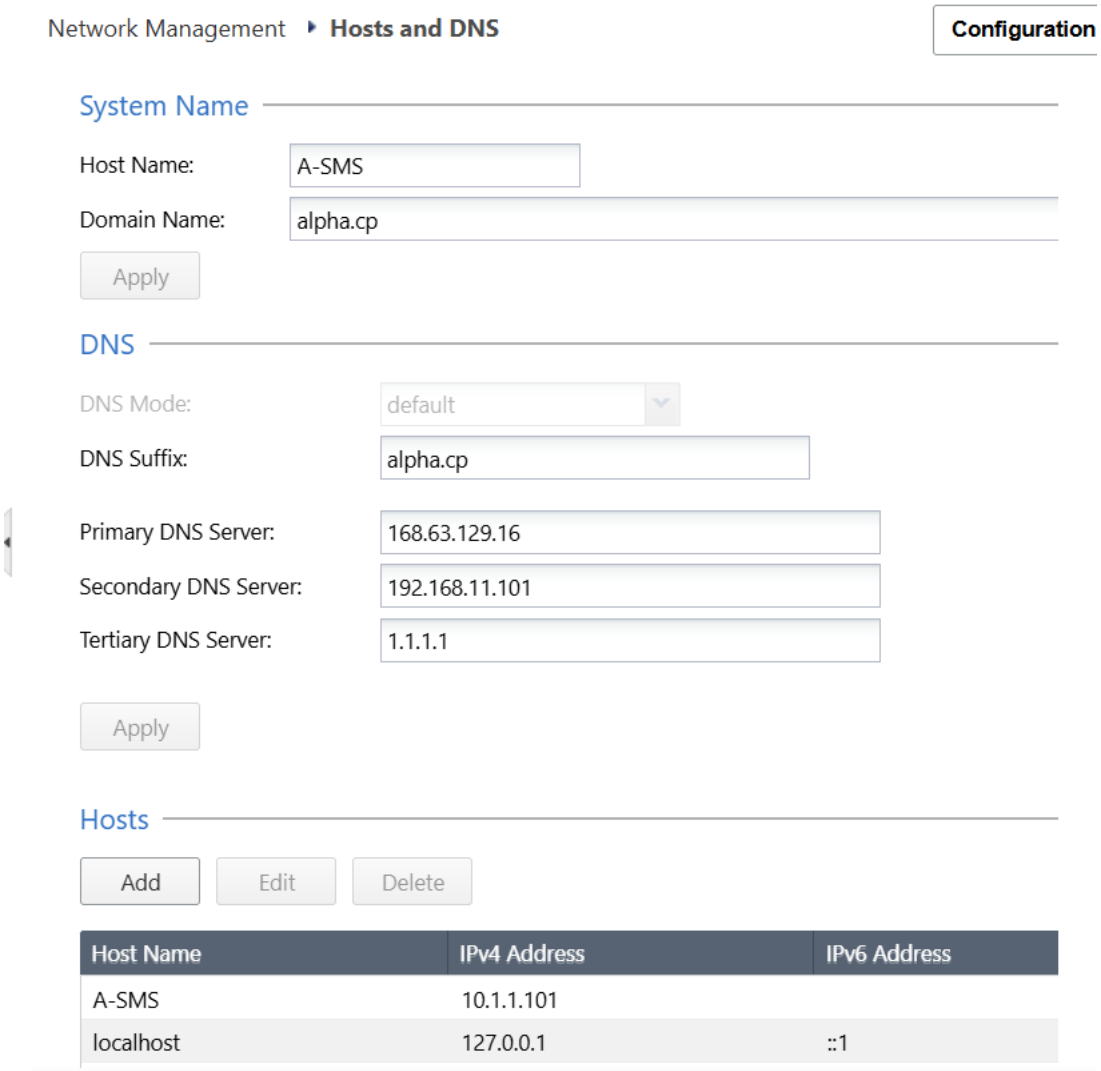


Fig. G4 DNS Resolver

C5: Locate Maintenance and Administration Functions.

Observation.

Function	Menu Path	Purpose
User Management	System Management → Users	Used to create, modify, and manage administrator accounts and permissions.
Time / NTP Settings	System Management → Time (or Time & Date)	Used to configure the system clock and synchronize time with NTP servers.
Maintenance (Backup/Snapshot)	Maintenance → System Backup / Snapshot Management	Used to create backups or snapshots for system recovery.

Difference Between a Configuration Backup and a System Snapshot

Configuration Backup

- Saves the system configuration and settings.
- Smaller in size and faster to create.
- Used to restore configuration after a failure or reinstallation.

System Snapshot

- Creates a complete image of the system, including the operating system, configuration, and installed software.
- Larger in size and takes longer to create.
- Used to restore the system to the exact state it was in when the snapshot was taken.

Conclusion:
I inspected The User Management, Time/NTP, and Maintenance pages. No backup, snapshot, update, or restore operation was performed.

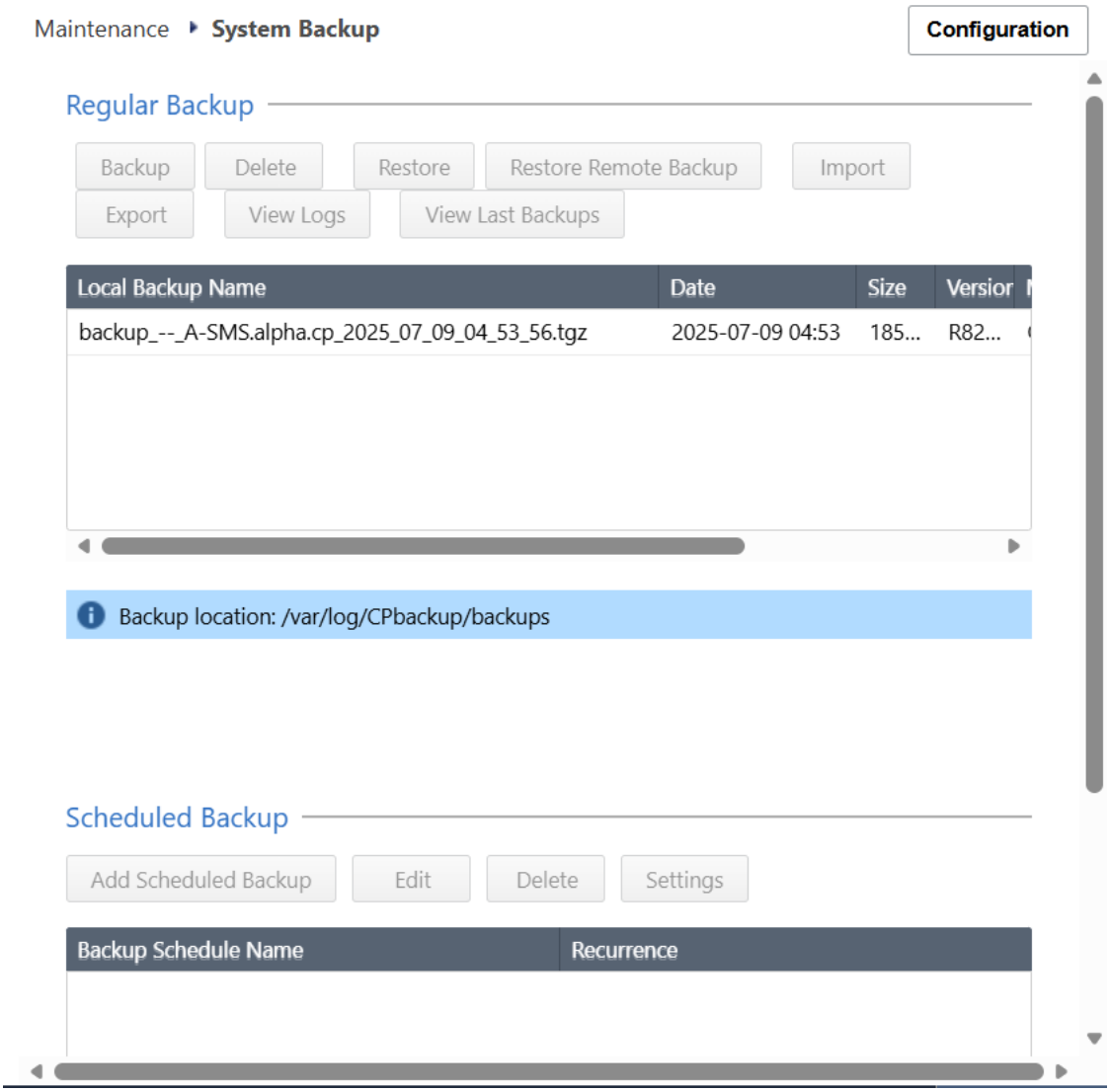


Fig. G5 screenshot of the backup page of A-SMS on Gaia portal

Part C Inspect Gaia Clish and Expert mode

C6: Collect system information in Clish

Observations:

1. show version all

What it reveals:

- Product release/version (e.g., R82)
- Build number
- Kernel version

- Platform information

Comparison with Gaia Portal: The information is the same, but Clish presents it as text output, while Gaia Portal presents it in a graphical interface

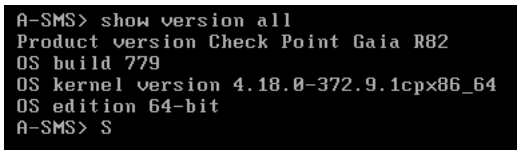


Fig. G6.1 Output of the command Show Version All

2. show uptime

What it reveals:

- How long the system has been running since the last reboot.
- Useful for verifying system stability and recent restart history.

Comparison with Gaia Portal: The uptime value is the same as that shown in the Gaia Portal, although a slight difference of a few seconds or minutes may occur because the information was checked at different times

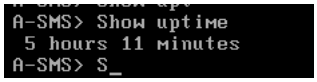


Fig. G6.2 Output of the Command Show Version All

3. show interfaces all

What it reveals:

- Details of all network interfaces.
- IP addresses, subnet masks, MAC addresses, interface status, and link state.

Comparison with Gaia Portal: The interface information matchs the Network Interfaces page in Gaia Portal. Clish displays all interfaces in a command-line format, while Gaia Portal displays them in a table and graphical view

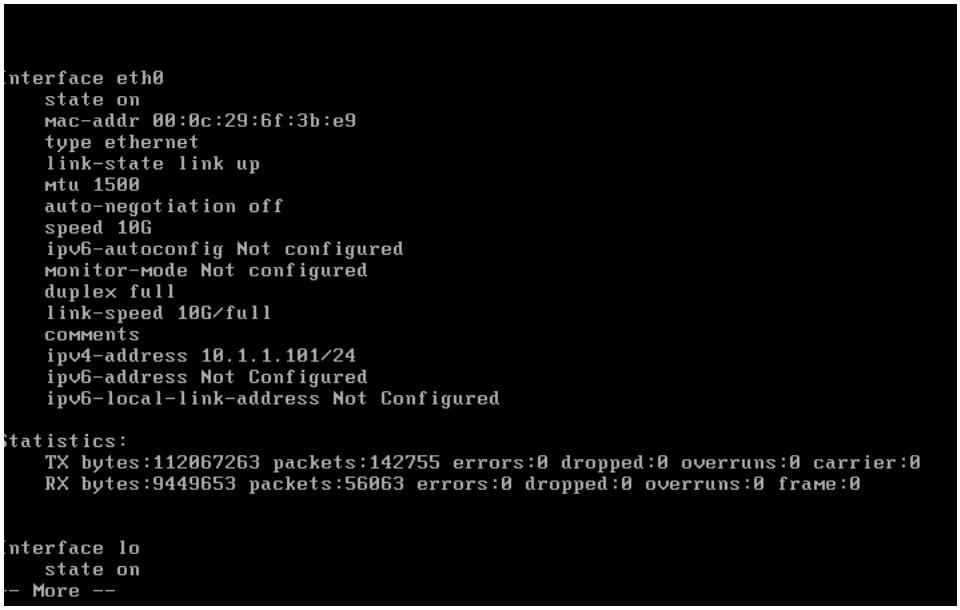


Fig G6.3 Output Of the Command Show interface All

4. show interface eth0

What it reveals:

- Detailed information for the eth0 interface only.
- IPv4 address, subnet mask, MAC address, administrative state, and link status.

Comparison with Gaia Portal: The eth0 IPv4 address and subnet mask matchs the values recorded in C3. This confirms that both management interfaces are reporting the same network configuration.

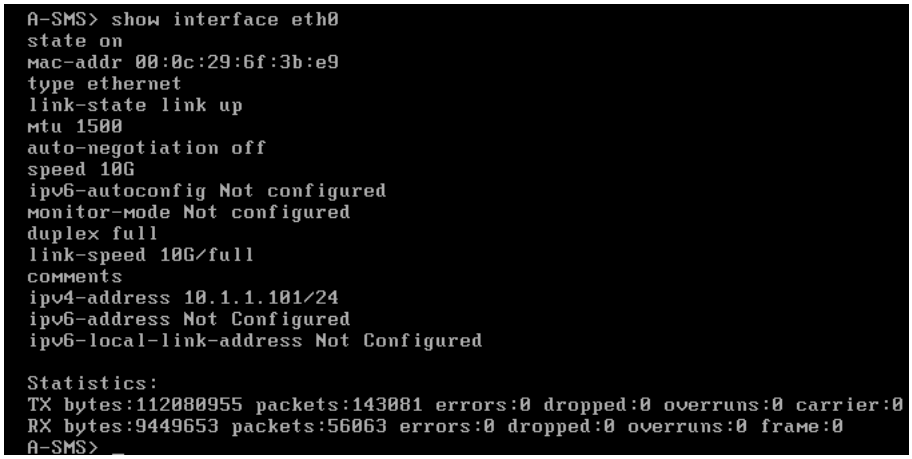


Fig. G6.4 Output of the command Show interface eth0

Difference Between Gaia Portal and Clish

- Gaia Portal: provides a graphical web-based interface that is easier to navigate and understand.

- Clish: provides a command-line interface that displays the same configuration information in text form and is often faster for administration and troubleshooting.

C7: Recognise the Expert shell

Command Results and Purpose

Command	Purpose
expert	Switches from Gaia Clish to the Expert shell.
whoami	Displays the current logged-in user, typically root.
pwd	Displays the current working directory, typically /home/admin or another Linux directory.
exit	Exits Expert mode and returns to Gaia Clish.

Difference Between Expert Mode and Clish

Gaia Clish

- Uses Check Point's structured command environment.
- Focused on system and network administration.
- Commands are limited to approved Gaia administrative functions.

Expert Mode

- Provides direct access to the underlying Linux operating system.
- Supports standard Linux commands and advanced troubleshooting tools.
- Offers greater administrative control and flexibility.

Prompt Change

When entering Expert mode, the command prompt changes from the Clish prompt to a Linux shell prompt, indicating that the administrator is now working in the operating system environment rather than the Gaia command environment.

Evidence

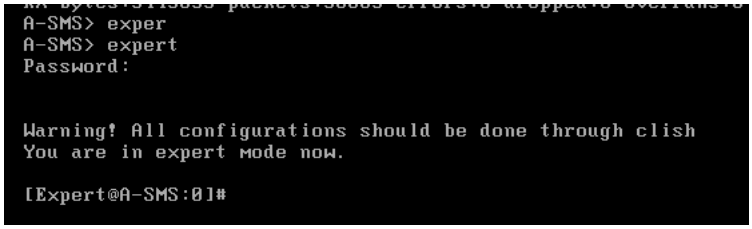


Fig.G7.1 Output of the command Expert (Switching to expert Mode)

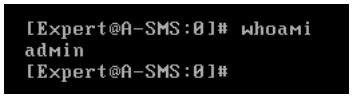


Fig. G7.2 Output of the command Whoami (displaying the current logged I user)

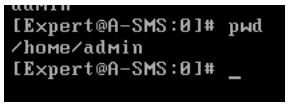


Fig. G7.3 Output of the command PWD (displaying the current directory I am)

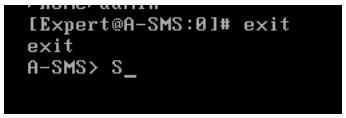


Fig. G7.4 Output of the command Exit (exiting the expert mode)

C8. Identify the Command Environment

Observation:
The Command Line panel in SmartConsole provides a way to run supported Check Point management commands from within the SmartConsole client.
Why ip route or ping -c may be rejected
Commands such as ip route and ping -c are Linux operating system commands, not SmartConsole management commands. The SmartConsole Command Line panel is designed for Check Point management and administrative commands, so some Linux commands may be unavailable or rejected.
Expert Mode provides direct access to the underlying Linux operating system of the Check Point server or gateway. ip route or ping -c are operating-system commands and should be run from an appropriate Expert session, where the full Linux command environment is available.

Part C: Navigate SmartConsole

C9: Inspect Gateways and Servers

Observations

Item	Value
Device Name	A-SMS
IP Address	10.1.1.101
Version	R82
Device Status	Active
Software Blades	Network Policy Management, Logging and Status
License Status	OK
Alerts	OK (currently empty)

Evidence:

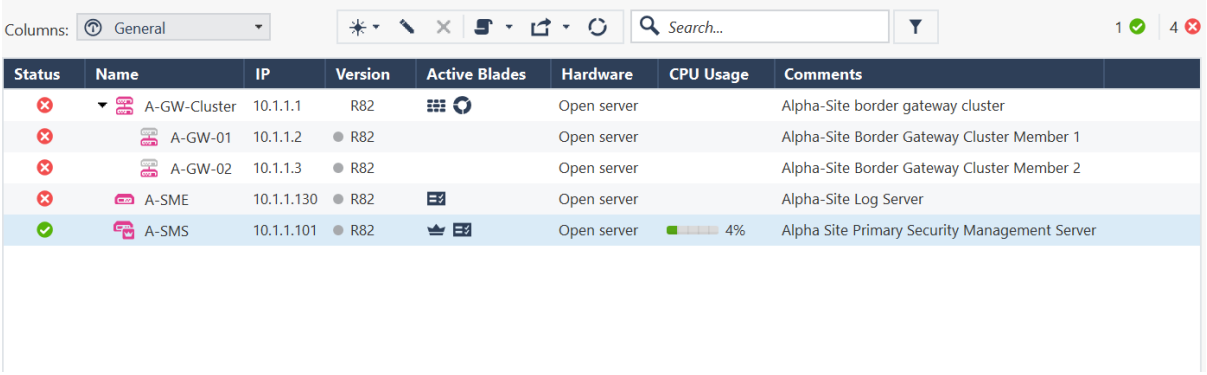


Fig. S1.1 A-SMS selected in Gateways & Servers

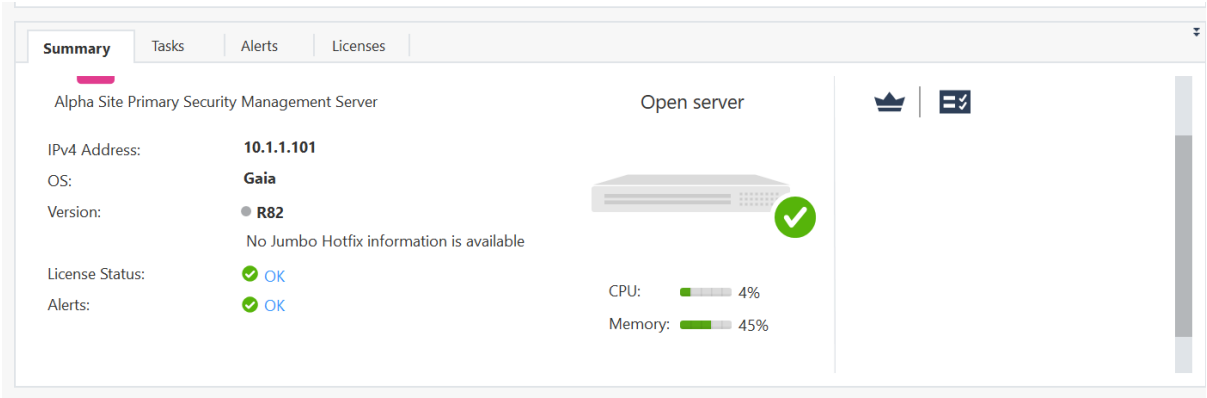


Fig. S1.2

The Summary page provides an overview of the Security Management Server, including its version, status, IP address, and enabled Software Blades.

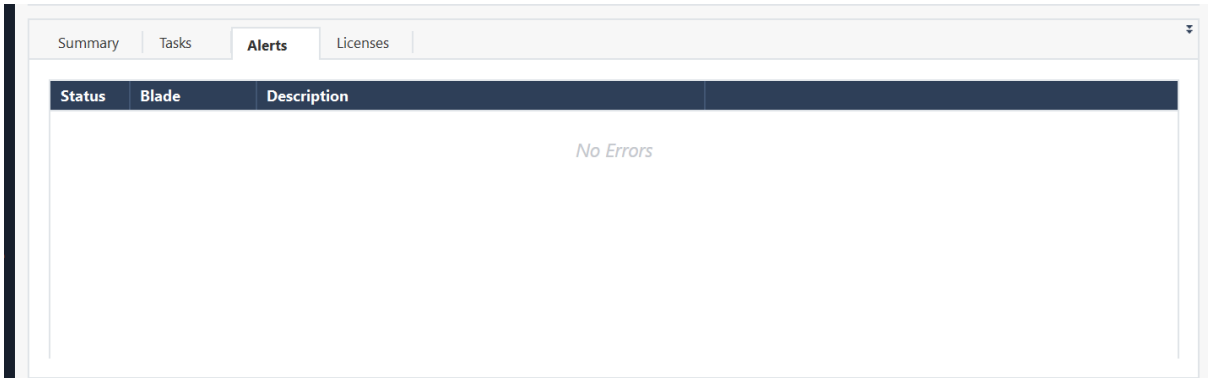


Fig. S1.3 Alerts (No any alert currently)

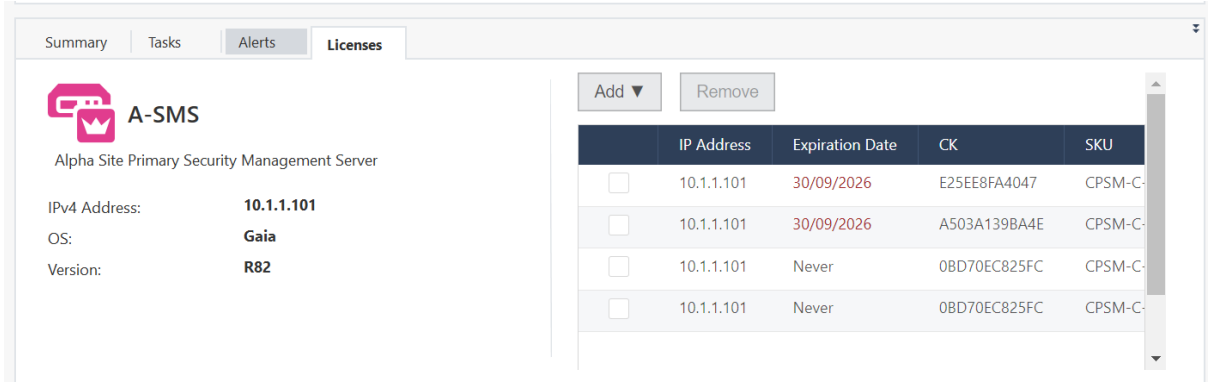


Fig. S1.4 License status (License is Ok)

Limitation Observed.

Only A-SMS is powered on during this lab. Therefore, any other preconfigured gateways or servers may appear unavailable, down, or unreachable. This does not necessarily indicate a fault; it reflects the lab environment where only the management server is running.

Conclusion.

The A-SMS object was inspected through the Gateways & Servers view. Device information, status, Software Blades, alerts, and license details were reviewed without modifying the configuration. Other preconfigured devices appeared unavailable because only the A-SMS virtual machine was powered on for this lab.

C10. Inspect an Existing Policy

Procedure:

Open Security Policies and locate the existing Access Control policy. Record the actual policy name displayed in your environment.

Field	Value
Policy Name	Policy No.1
Rule Name	Cleanup rule
Source	Any
Destination	Any
Source and Applications	Any
Action	Accept
Task	Log

Explanation:
This rule permits Any traffic from any source to any destination. The traffic is accepted by the gateway, and each connection is recorded in the logs for monitoring purposes.

Evidence:

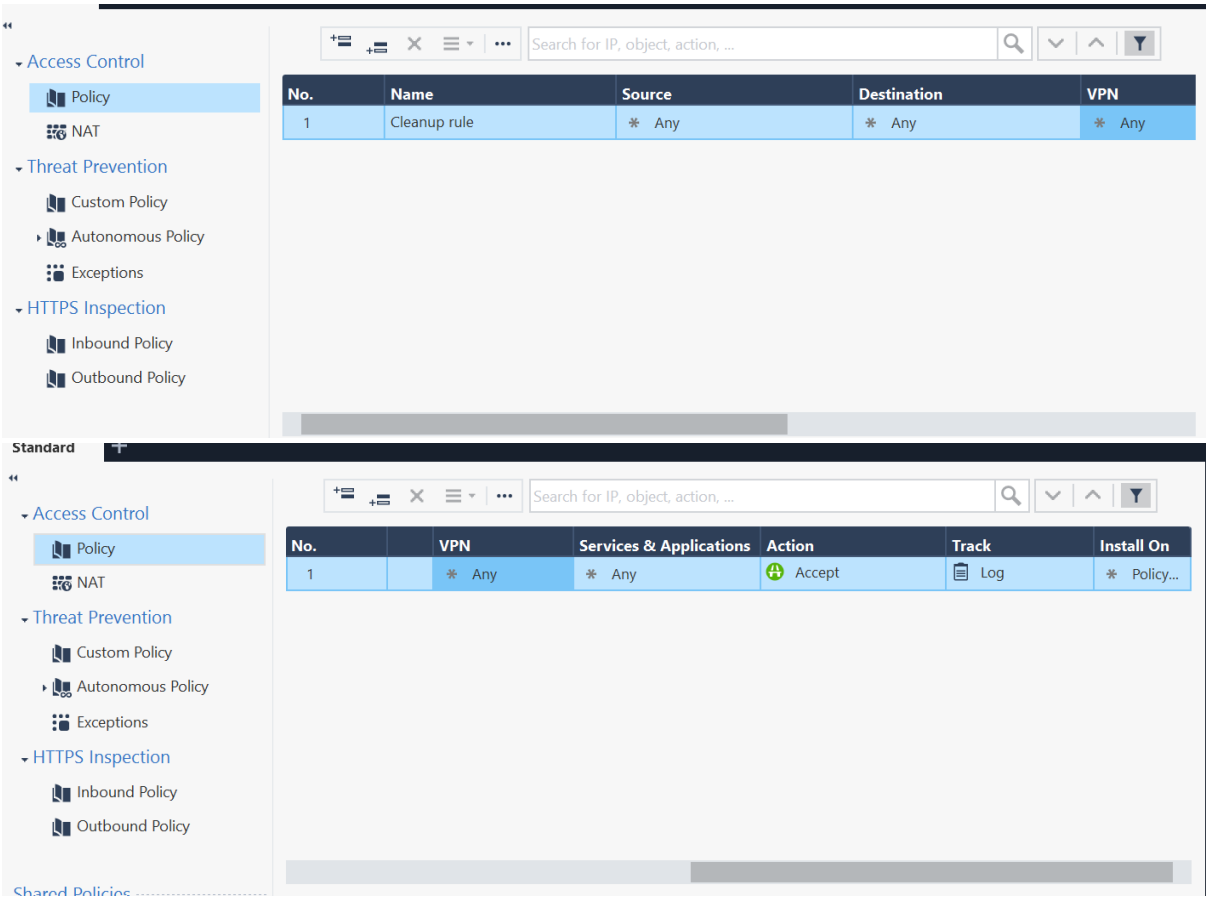


Fig. S2 Access Control Policy

C11. Explore Objects and Administrative Controls

Objects Inspection

Object	Type	Address/Identifier	Property showing it Purpose
A-SMS	Security Management Server	10.1.1.101	Hosts the Check Point management services and manages security policies.
A-LDAP	Host Object	192.168.11.101	It is the Alpha-Site Domain Controller and File Server, showing its role in authentication and file services.

Administrative Controls

Area	Purpose
Manage & Settings	Used to manage administrators, permissions, system settings, and SmartConsole configuration.
Session Controls	Used to manage the current session, including viewing changes, publishing changes, or discarding changes.
Tasks Window	Displays the progress and status of administrative tasks such as policy installations and other background operations.

Evidence:

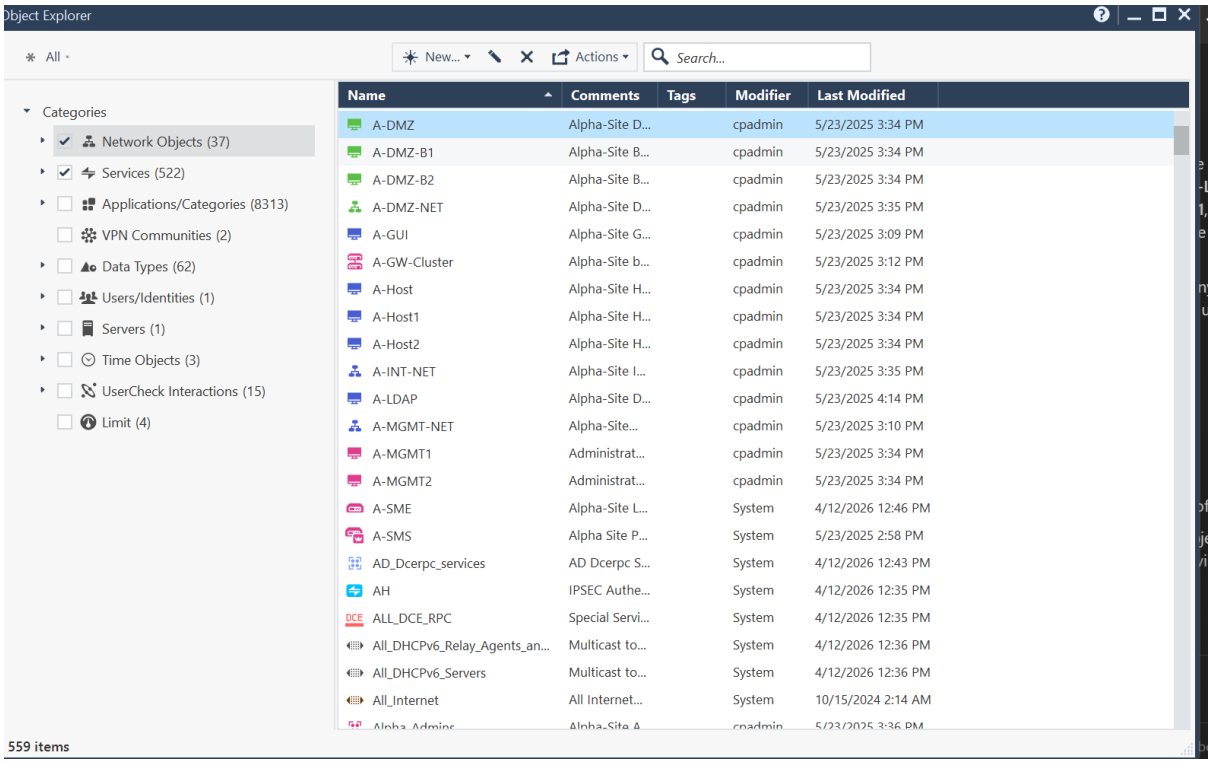


Fig S3.1 Object Explorer with A-SMS

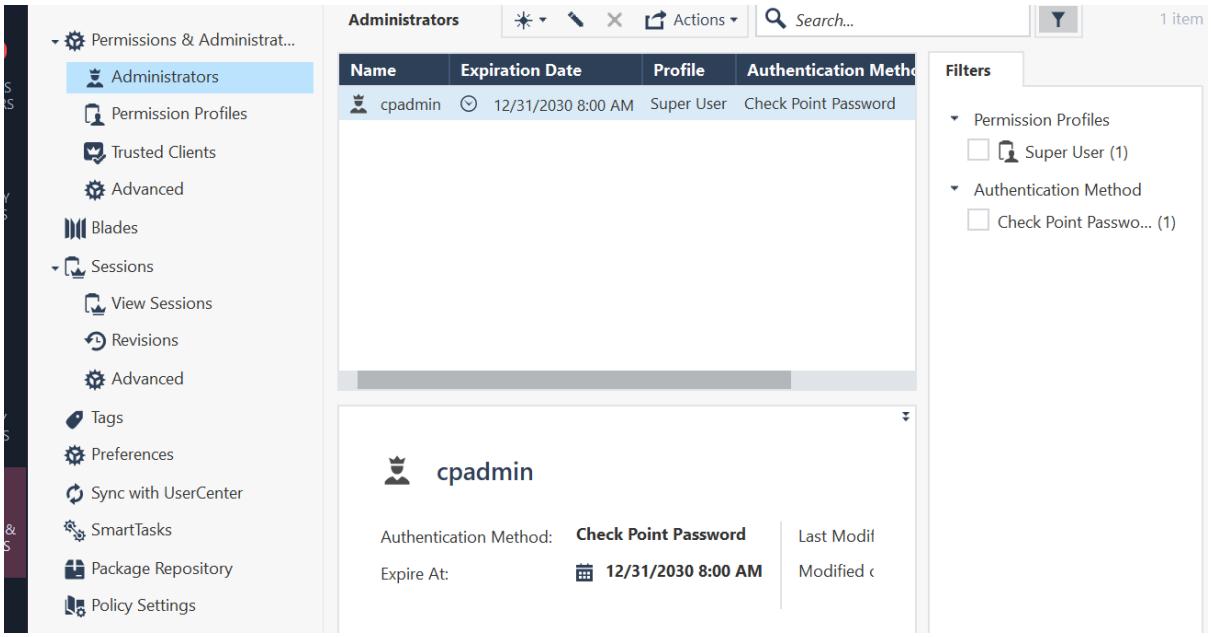


Fig. S3.2 Manage & Settings

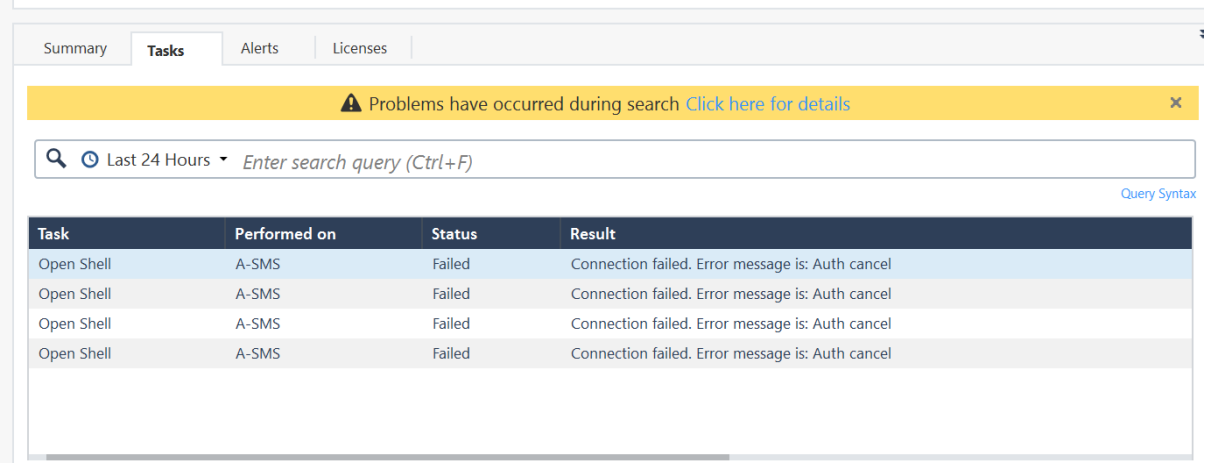


Fig. S3.3 Task window

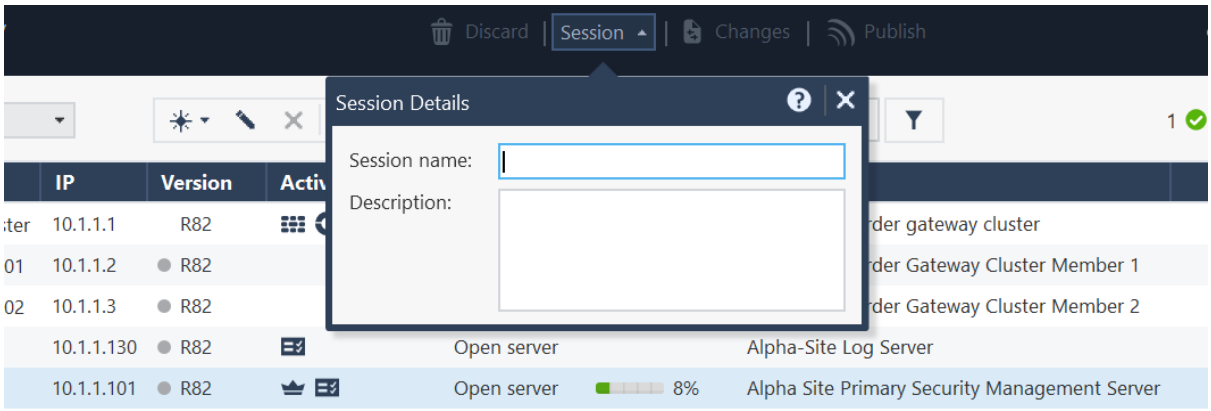


Fig. 3.4 Session Control

Observation

The Object Explorer was used to inspect A-SMS and another existing object. Session Controls and the Tasks window were also located, demonstrating how SmartConsole manages administrative changes and background operations.

C12. Inspect Logs and Events

Item	Observation
Log Source / Origin	A-SMS
Time Range	Last 24 Hours
Meaningful Detail	The Logs & Events view returned at least 63 results, indicating that log data is being collected by A-SMS.

Explanation:

The selected log source was A-SMS, and the time range was set to Last 24 Hours. The log view displayed multiple entries, with the most recent visible entry recorded at 7:00:00 AM. The results indicate that the management server is receiving and storing log information.

The Action field is not visible in the current screenshot, so it cannot be accurately determined without opening a log entry or displaying the Action column

Observation:

The Logs & Events view showed log entries originating from A-SMS with the time range set to Last 24 Hours. At least 63 log records were available, demonstrating that logging was active and log data was being collected by the management server

Evidence:

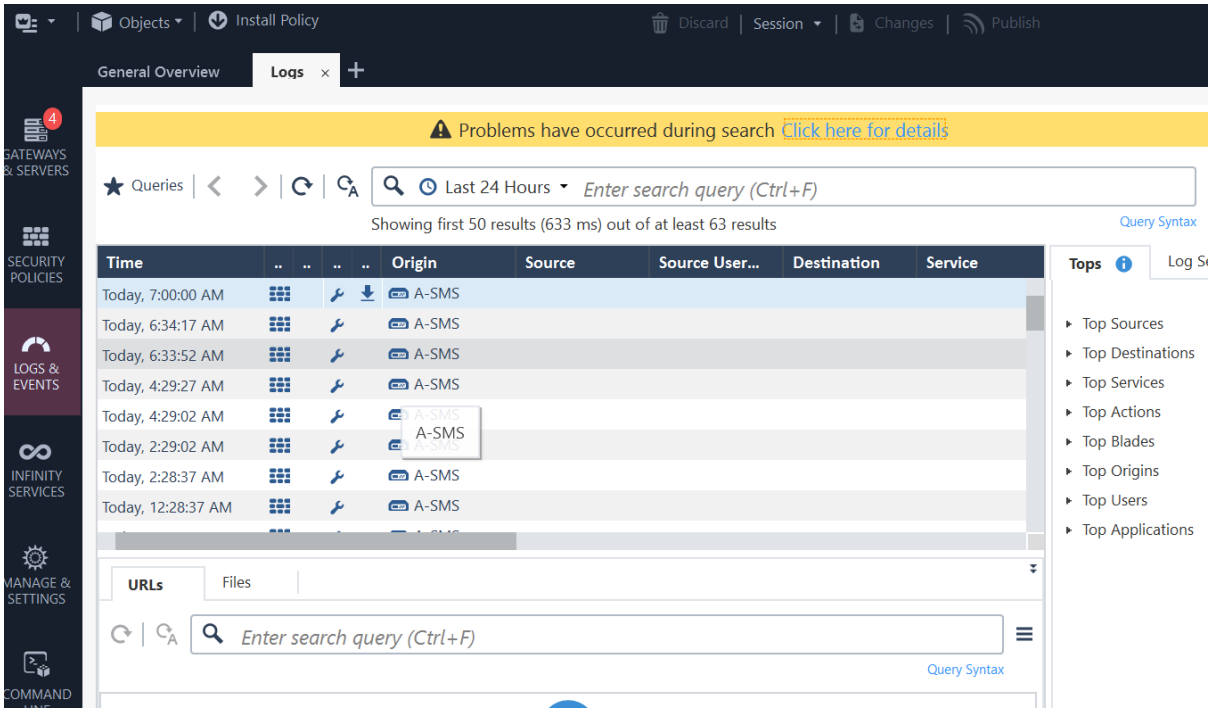


Fig. S4 Log entries from the A-SMS server

C13. Locate Publication and Installation Controls

The Publish and Install Policy controls were located in the SmartConsole toolbar at the top of the window**.**

Control	Item
Publish	Saves and commits configuration changes made during the current

	session to the Management Server database, making them visible to other administrators
Install Policy	Deploys the published security policy from the Management Server to the Security Gateway, where the rules are enforced.

Relationship to B4

As explained in B4, publishing and policy installation are separate actions. Publishing saves administrative changes, while installing a policy applies those changes to the gateway. This separation allows changes to be reviewed before deployment.

Command Line Panel

The Command Line panel was located in the left navigation menu of SmartConsole.

Interface	Purpose
SmartConsole Command Line Panel	Used for Check Point management-related commands from within SmartConsole.
Gaia Clish	Structured administration shell used for system and network configuration.
Expert Mode	Full Linux shell used for advanced administration, diagnostics, and troubleshooting.

Unlike Clish and Expert Mode, the SmartConsole Command Line panel is part of the management application and may not support Linux operating-system commands such as ip route or ping -c.

Observation:

The Publish and Install Policy controls were observed but not used. The Command Line panel was also located and inspected. No changes were published, no policy was installed, and no commands were executed from the SmartConsole Command Line panel. Any inaccessible pages or objects would need additional verification before conclusions could be drawn about their status or configuration.